

Mohammad Ful Hossain Seikh

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Objective: To develop concepts and skills in experimental Astroparticle Physics and also to gain some experience from cutting edge researches in the best laboratories around the world.

Educational Qualifications:

Degree	Years	Institution	Performance
Ph.D. in Physics	2020-Present	University of Kansas, USA	3.98/4.0
M.S. in Electrical Engineering	2024-Present	University of Kansas, USA	4.0/4.0
M.Sc. in Physics	2018-2020	Aligarh Muslim University, India	9.170/10 (Gold Medalist)
B.Sc. Honors in Physics	2015-2018	Aligarh Muslim University, India	9.186/10 (Gold Medalist)
Higher Secondary (12th)	2014	CBSE, India	9.120/10 (First Position)
Secondary (10th)	2012	CBSE, India	10/10 (Certificate of Merit)

Research Experiences:

- Currently working on the **Data Analysis of Askaryan Radio Array (ARA) Station 1** in search for ultra-high energy (UHE) neutrino under the supervision of Professor David Besson, the University of Kansas (KU), USA.
- Monitored **IceCube Neutrino Observatory, South Pole** on behalf of KU and presented weekly reports in their Monitoring calls (21st-28th June, 2021; 4th-11th April, 2022; 11th-16th June, 2023; 24th-30th June, 2024; 20th-26th January, 2025).
- Worked on master's thesis and wrote a review report titled **Light Sterile Neutrinos: Status and Prospects** under the supervision of Prof. Mohd. Sajjad Athar, Aligarh Muslim University (AMU), India (June, 2020) (Researchgate DOI: 10.13140/RG.2.2.13777.66405).
- Wrote a manuscript titled **Theoretical Determination of the Triangular Neutrino Oscillations Parameters: A New Approach** and gave a presentation on it at the 7th IAPT (Indian Association of Physics Teachers) National Students' Symposium of Physics conducted at Panjab University (3rd-6th October, 2019).
- Was selected for Summer Students' Visiting Program-2019 and worked on a project titled **Importance of 1-3 Mixing Angle in Three Flavor Neutrino Oscillations Paradigm** at Institute of Physics, Bhubaneswar, in summer (May 16, 2019 - July 24, 2019) under the supervision of Dr. Sanjib Kumar Agarwalla (Researchgate DOI: 10.13140/RG.2.2.34421.12009).
- Worked on **The Weak Interaction Physics** as a winter intern at Indian Institute of Technology-Banaras Hindu University (IIT-BHU), India under the supervision of Dr. Gauhar Abbas (December, 2019 - January, 2020).
- Worked on a project titled **Quarks and Leptons**, where I made a detailed report on the history of the development of particle physics and performed different studies on Standard model interactions under the supervision of Prof. Mohd. Sajjad Athar, AMU (2017).

Publications:

- Full list of publications can be found on Inspire HEP, Google Scholar, ResearchGate. This includes several ARA, RNO-G, ARIANNA, IceCube and RET collaboration publications for which I am listed as an author, but was not either a primary or secondary author.
- **M.F.H. Seikh***, "Physics of radio antennas", European Physics Journal-Special Topics, 2024, arXiv:2411.07507.

- **M.F.H. Seikh* for the ARA Collaboration**, “Askaryan Radio Array: searching for the highest energy neutrinos”, European Physics Journal-Special Topics, 2024, arXiv:2411.01761.
- **M.F.H. Seikh* and D. Besson**, “IceCube Neutrino Observatory: A Cold Discovery Hub”, Physics Bulletin - February 2024, Department of Physics, AMU.
- **M.F.H. Seikh* and D. Besson for the ARA Collaboration** “Calibration and Physics with ARA Station 1: A Unique Askaryan Radio Array Detector”, PoS(ICRC2023)1163, arXiv:2308.07292.
- **M.F.H. Seikh***, “Neutrinos and the 40 Future Routes”, Physics Bulletin - February 2020, Department of Physics, AMU Researchgate DOI: 10.13140/RG.2.2.27981.10729.

Editorial & Reviewer Experience:

- Working in the editorial board (M.S. Athar, D. Besson, **M.F.H. Seikh**, V. Grachev, I. Karvchenko) of European Physics Journal Special Topics (EPJ-ST) journal for a book titled “Radio Detection of Ultra-high Energy Neutrino & Cosmic Ray” with 10+ articles.
- Reviewed 9 scientific articles for the Springer’s EPJ-ST journal.
- Started an e-Magazine, **The Magnetic Monopole** and edited its first Issue published by Society of Physics Students, KU (June 2022).

Computation Skills:

- **Programming Skills:** C/C++, CERN ROOT, AraRoot, Python, FORTRAN.
- **Software Skills:** Wolfram Mathematica, HFSS.
- **Operating System:** Linux (Ubuntu), Windows, MacOS.
- **Documenting:** L^AT_EX, Microsoft Office.

Awards & Achievements:

- Awarded **2025-26 Ralston Dissertation Fellowship** by the Department of Physics & Astronomy, KU, USA.
- Awarded **2025 McKeithan Summer Fellowship** by the Department of Physics & Astronomy, KU, USA.
- Awarded **the 2024-2025 Graduate Student Award for Distinguished Service** by the Office of Graduate Studies, KU, USA.
- Awarded **Graduate Student Travel Fund** from the Department of Physics & Astronomy, KU, USA, for presenting an invited seminar talk at the Physics Department, Ohio State University, January 2025.
- Awarded **Wallace S Strobel Scholarship 2024-2025** from the EECS Department, KU, USA.
- Awarded **2024 Ralston Summer Fellowship** from the Department of Physics & Astronomy, KU, USA.
- Awarded **DAP Travel Grant 2024** and **DPF Travel Award 2024** from American Physical Society (APS) for APS April Meeting 2024 at Sacramento, CA.
- Awarded **Doctoral Student Research Fund** from the Office of Graduate Studies, KU, USA, for the possible field trip to the NSF’s Summit Station, Greenland in Summer 2024.
- Awarded **Graduate Student Travel Fund** from the Department of Physics & Astronomy, KU, USA, for ICRC 2023, Nagoya, Japan.

- Awarded **Distinguished Students Award 2021-2022** and **DPF Travel Grant 2022** from American Physical Society (APS) for APS April Meeting 2022 at New York City, NY.
- Awarded **CLAS Graduate Scholarly Development Fund 2022** from KU for RNO-G Fall Collaboration Meeting at Chicago.
- Awarded **Carl & Catherine Chaffee Scholarship 2020-2021** from the Department of Physics and Astronomy, KU, USA.
- Awarded **University Gold Medal** during Master's degree (M.Sc.) among 50 students in the Department of Physics, AMU, India (2020).
- Awarded **Undergraduate Topper Scholarship** by the University Grants Commission for securing First position in Bachelor of Science, (February 2020).
- Awarded **University Merit Financial Award** by the Department of Physics, AMU, India for 2 years (2019-2020).
- Awarded **University Gold Medal** for Highest % of marks in B.Sc. (Hons) in the Faculty of Science, AMU, India (2018).
- Awarded **University Gold Medal** during Bachelor's degree (B.Sc.) among 120 students in the Department of Physics, AMU, India (2018).
- Awarded **Abdul Aziz Medal** during B.Sc. in the Department of Physics, AMU, India (2018).
- Awarded **Washington DCs Scholarship** for five consecutive years (2015-2019) by Aligarh Alumni Association.
- Awarded **Sir Syed Global Scholar Award (SSGSA)** by 'Sir Syed Education Society of North America', an American organization that selects emerging scholars from India in 2018.
- Secured **First Position** among 40 students in the higher secondary exam (12th).
- Awarded **Certificate of Merit** for obtaining the top Grade (CGPA 10/10) in all five subjects in High School (10th).

Outreach / Broader Impact:

- Hosted a first of its kind workshop in India on **Radio Neutrino & Cosmic Ray Astronomy** at "Aligarh Muslim University, India" (18th-20th April, 2024).
- Conducted an Outreach event on 'Internships & Training Opportunities for Physics Research' for the students at AMU (October 17, 2021).
- Participated in the Science Academies' **Science Leadership Workshop** (India's First Science Leadership Program) and scored 10/10 in the exit exam organized by the Central University of Punjab, Bathinda, India (June 22-28, 2020).
- Participated and scored 90% marks in "General Awareness Quiz" organized under IQAC, JNS Govt College, Shujalpur, India, during the lockdown period (June 4, 2020).
- Worked as a member of SSGSA's AMU Campus Team during the session 2019-20.
- Volunteered in **One Person One Tree** (A Mass Tree Plantation Drive) in AMU (April 25, 2017).
- Participated in **Blood Donation Camp** as a DONOR held by AMU Student Union (March 25, 2017).

Independent Courseworks:

- **Responsible Conduct of Research (RCR) Basic Course** by **CITI Program**. Certificate earned on Wednesday, October 6, 2021, Grade Achieved: 100%, Record ID 45387027.

- **COVID-19: Public Training Series - Insights for Higher Ed Leaders** by CITI Program. Certificate earned on Wednesday, September 29, 2021, Grade Achieved: 100%, Record ID 45248931.
- **COVID-19: Public Training Series - Back to Campus (Fall 2020)** by CITI Program. Certificate earned on Wednesday, September 29, 2021, Grade Achieved: 100%, Record ID 45248928.
- **From the Big Bang to Dark Energy** by the University of Tokyo on Coursera. Certificate earned on Thursday, May 23, 2020, Grade Achieved: 100.0%.
- **Particle Physics: an Introduction** by the University of Geneva on Coursera. Certificate earned on Thursday, June 6, 2019, 1:00 PM GMT, Grade Achieved: 100.0%.
- **“Senior Diploma in Painting”** from Bangiya Sangeet Parishad, West Bengal, India.

Invited Lectures/Talks:

- Presented an invited talk on **Radio Engineering for Particle Astrophysics Experiments** at the HEP seminar at the University of Nebraska - Lincoln (March 26, 2025).
- Presented an invited talk on **Radio Engineering for Particle Astrophysics Experiments** in the Astroparticle seminar at the Ohio State University (22nd January, 2025).
- Hosted a mini-workshop and lecture series on **Radio Engineering & Antenna Physics** at the “IEI Summer School 2024, University of Alaska, Anchorage, USA” (2nd-5th July, 2024).
- Presented an invited talk on **Ultra-high Energy Neutrino & Cosmic Ray Detectors** at the “ICEHAP Seminar, Chiba University, Japan” (9th August, 2023).
- Hosted a mini-workshop and lecture series on **Antenna Physics** at the “IEI Summer School 2023, South Dakota School of Mines and Technology, USA” (2nd-5th July, 2023).
- Presented an invited talk on **Ultra-high Energy Neutrino & Cosmic Ray Detectors** at the “Physics Colloquium, AMU, India” (12th January, 2023).
- Presented an invited talk on **Antennas in Ultra-high Energy Neutrino & Cosmic Ray Detectors** at the “Particle Physics Seminar, University of Delaware” (4th October, 2022).

Conferences / Workshops / Seminars:

- Presented two talks on **RNO-G in-ice Antennas** in the “RNO-G Collaboration Meeting 2024, Upsala University” (9th-14th September, 2024).
- Attended the “ICTS Neutrino School 2024” and presented a poster on **Overview of Askaryan Radio Array Detector** at the “International Center for Theoretical Sciences, India” (22nd April-3rd May, 2024).
- Presented a talk on **Radio Frequency Instrumentation** at the “Radio Neutrino & Cosmic Ray Astronomy Workshop, AMU, India” (18th-20th April, 2024).
- Presented a talk on **Towards a Five Station Analysis of Askaryan Radio Array** “APS April Meeting, Sacramento” (3rd-6th April, 2024).
- Presented a talk on **ARA Antenna Models** at the “ARA Collaboration Meeting, Ohio State University, Columbus” (5th-8th March, 2024).
- Presented a talk on **Current Status & Future Prospects of Askaryan Radio Array, South Pole** at the “APS Prairie Section Meeting, University of Missouri, Columbia” (30th November-2nd December, 2023).
- Presented a talk on **Introduction to Antenna Physics** at the “RET Collaboration Meeting, Vrije University Brussels, Belgium” (8th-11th November, 2023).

- Presented a talk on **Gain Calibration of RNO-G Surface LPDA Antennas** at the “Fall Collaboration Meeting, Penn State University” (11th-15th September, 2023).
- Presented a poster on **Calibration and Physics with ARA Station 1: A Unique Askaryan Radio Array Detector** at the “38th International Cosmic Ray Conference (ICRC 2023), Nagoya University, Japan” (29th July, 2023).
- Presented a talk on **Galactic Noise From The Milky Way & Its Use As A Diagnostic Tool For Radio Neutrino Experiments** in the “Astro Seminar, KU” (28th October, 2022).
- Presented a talk on **Towards the Calibration of Askaryan Radio Array (ARA) Station 1** in the “2022 APS April Meeting” in New York City (9th-12th April, 2022) 2022APS..APRZ03001S.
- Presented a talk on **Ice Calibration Planning for 2023 Season** in the “RNO-G Collaboration Meeting 2022, University of Chicago” (8th April, 2022).
- Participated and presented a paper titled **Theoretical Determination of the Triangular Neutrino Oscillations Parameters: A New Approach** in the “VirtCon2020” jointly organized by Department of Physics, Bhairab Ganguly College in association with Department of Physics, Dum Dum Motijheel College (6th and 7th July, 2020).
- Participated in the ten days Online workshop on **Python for a Cause: Online Workshop on Python Computing** organized by Teach Python in association with Indian Institute of Science Education and Research (IISER) Kolkata (12th-24th June, 2020).
- Attended a two days training program on **Quadcopter/Drone** at Zakir Husain College of Engineering and Technology, AMU (3rd-4th September, 2016).
- Attended and Participated in **Calligraphy Workshop** at Cultural and Educational Center lawn, AMU (23rd and 24th December, 2015).
- Participated in **Regional Science Congress 2012** at Jawahar Navodaya Vidyalaya, Durgapur, India (26th November-1st December, 2012).
- Participated at Navodaya Vidyalaya Samity **Cluster Level in Science: Biology Exhibition** and Secured 2nd Position at Jawahar Navodaya Vidyalaya, Gopalpur, India (24th-25th September, 2012).
- Attended the lecture series/workshop on **An Introduction to Tensor Analysis and its Applications** delivered by Prof. Zafar Ahsan organized by AMU Society of Physics (19th-28th February, 2018).

Professional Affiliations:

- Graduate Students Representative - Member at Large, Physics & Astronomy Graduate Student Organization (P&A GSO), KU, USA, 2025-present.
- Graduate Students Representative Liaison, P&A GSO, KU, USA, 2023-2024.
- Executive Officer, P&A GSO, the KU, USA, 2023-2024.
- Graduate Student Member, APS, USA, 2020-Present.
- Team Lead, Promotions & Outreach Team, SSGSA 2023-Present.
- Graduate President, Society of Physics Students (SPS), KU Chapter 2021-2022.
- Lifetime Member, Sigma Pi Sigma, the National Physics Honor Society.
- Student Member, National Society of Black Physicist, USA 2021-2022.
- Member, Mentorship & Applications Team, SSGSA 2021-2022.
- Visiting Winter Trainee, Indian Institute of Technology, BHU, India, 2019-2020.
- Visiting Summer Student, Institute of Physics, Bhubaneswar, India, 2019-2020.
- Member, AMU Society of Physics (AMUSoP), India, 2018-2020.